Chapter 8

Quasi-Likelihood and Estimating Equations

[Quasi-Likelihood and Estimating Equations]

8.1 Quasi-Likelihood

8.1.1 Connection with Exponential Dispersion Families

To motivate quasi-likelihoods, consider maximum likelihood estimation of generalized linear models. In deriving derivatives of the log likelihood for these models we made use first of independence among response variables to write the likelihood and its derivatives in terms of sums of contributions from individual variables Y_i, and then also used the chain rule. Consider using these same techniques, but taking the derivation only to the point of obtaining a derivative of the log likelihood with respect to the expectation

μ_i,

$$\frac{\partial \ell_i(\mu_i, \phi)}{\partial \mu_i} = \frac{\partial \ell_i(\mu_i, \phi)}{\partial \theta_i} \frac{d\theta_i}{d\mu_i}.$$

$$\frac{\partial \ell(\boldsymbol{\mu}, \phi)}{\partial \boldsymbol{\mu}} = \sum_{i=1}^{n} \frac{\partial \ell_i(\mu_i, \phi)}{\partial \mu_i}.$$

An individual response variable has an exponential dispersion family form for its density or mass function,

$$f(y_i|\theta_i, \phi) = \exp[\phi\{y_i\theta_i - b(\theta_i)\} + c(y_i, \phi)].$$

As for our discussion of basic generalized linear models in Stat 520, be aware that different authors use different forms for the dispersion parameter – my ϕ may be $1/\phi$ for another author, and both of these may be $1/a(\phi)$ for yet another author. With this form, we have $E(Y_i) = \mu_i = b'(\theta_i)$ and $var(Y_i) = (1/\phi)b''(\theta_i) = (1/\phi)V(\mu_i)$. Differentiating the logarithm of this density with respect to μ_i we obtain,

$$\frac{\partial \ell_i(\mu_i, \phi)}{\partial \mu_i} = \frac{\partial \ell_i(\mu_i, \phi)}{\partial \theta_i} \frac{d\theta_i}{d\mu_i} = \frac{\phi\{y_i - b'(\theta_i)\}}{b''(\theta)} = \frac{\phi\{y_i - \mu_i\}}{V(\mu_i)}. \tag{8.1}$$

Notice that expression (8.1) (and thus also the sum across i) depends only on the first two moments of Y_i, $E(Y_i) = \mu_i$ and $var(Y_i) = (1/\phi)V(\mu_i)$.

Now consider a situation in which we know that the left most term in (8.1) is equal to the rightmost term for some function $V(\cdot)$, but without the knowledge that the response distributions are exponential dispersion family. Could we "recover" the log likelihood for μ_i (for fixed ϕ) through integration? Consider

$$Q_i(\mu_i|\phi) = \int_{y_i}^{\mu_i} \frac{\phi\{y_i - t\}}{V(t)}\, dt,$$

$$Q(\boldsymbol{\mu}|\phi) = \sum_{i=1}^{n} \int_{y_i}^{\mu_i} \frac{\phi\{y_i - t\}}{V(t)}\, dt. \tag{8.2}$$

The limits of integration in (8.2) are chosen based on what we hope to obtain, which is a function that is $\ell_i(\mu_i, \phi)$ up to an additive constant. Since the integrand is the derivative of $\ell_i(\mu_i, \phi)$ with respect to μ_i, the upper limit is obvious. The lower limit is to obtain a function of y_i alone, that then becomes an additive constant for the log likelihood.

Example 8.1

Suppose $Y_1, \ldots, Y_n$ have Poisson distributions with expected values $\mu_1, \ldots, \mu_n$, but we are not aware of this fact. We are, however, given that,

$$\frac{\partial \ell(\mu_i)}{\partial \mu_i} = \frac{y_i - \mu_i}{\mu_i}.$$

Applying (8.2),

$$Q_i(\mu) = \int_{y_i}^{\mu_i} \frac{y_i - t}{t}\, dt = y_i \log(\mu_i) - \mu_i, +k(y_i)$$

so that,

$$Q(\boldsymbol{\mu}) = \sum_{i=1}^{n} \{y_i \log(\mu_i) - \mu_i\} + K(\boldsymbol{y}).$$

Now, if $Y_i \sim$ iid $\text{Po}(\mu_i)$ for $i = 1, \ldots, n$,

$$\ell(\boldsymbol{\mu}) = \{y_i \log(\mu_i) - \mu_i\} + w(\boldsymbol{y})$$

and, up to an additive constant, $Q(\boldsymbol{\mu}) = \ell(\boldsymbol{\mu})$.

Note that, in general, the additive constant that distinguishes $Q(\boldsymbol{\mu})$ from $\ell(\boldsymbol{\mu})$ in this situation (which is independent response variables with exponential dispersion family distributions) depends on both $\boldsymbol{y}$ and ϕ; this will be important if the value of ϕ is to be estimated.

Now, nothing in this introductory section is anything we do in practice. Its purpose was simply to show a connection between quasi-likelihood and generalized linear models, and to motivate the notion that considering the quantity (8.2) makes some sense.

8.1.2 Basic Quasi-Likelihood

The fundamental idea underlying basic quasi-likelihood is that, even in situations for which we do not have independent random variables with fully specified exponential dispersion family distributions, the function $Q(\boldsymbol{\mu}|\phi)$ in (8.2) should behave in a manner that resembles a log likelihood function for $\boldsymbol{\mu}$.

Consider independent random variables $Y_1, \ldots, Y_n$ that have expectations $\mu_1, \ldots, \mu_n$ and variances $(1/\phi)\, V_1(\mu_1), \ldots, (1/\phi)\, V_n(\mu_n)$ for a set of specified functions $V_1(\cdot), \ldots, V_n(\cdot)$. Assume that $\mu_i = h(\boldsymbol{x}_i, \boldsymbol{\beta})$ for some known function $h(\cdot)$ and unknown parameters $\boldsymbol{\beta}$ with $\dim(\boldsymbol{\beta}) = p < n$, but specify nothing additional about the model. Notice that we have allowed the functions $V_i(\cdot)$ to vary across observations. In the majority of situations it will be reasonable to take these to be the same function, but that is not necessary. What is necessary, however, is that $var(Y_i) = (1/\phi)\, V_i(\mu_i)$ where ϕ is constant and $V_i(\cdot)$ does not depend on elements of $\boldsymbol{\mu}$ other than μ_i (McCullagh and Nelder 1989, p. 324).

In this independence situation, take the quasi-likelihood to be as given in (8.2). The quasi-score function is then,

$$\begin{aligned} U_i(\mu_i|\phi) &= \frac{\phi\{y_i - \mu_i\}}{V_i(\mu_i)} \\ U(\boldsymbol{\mu}|\phi) &= \sum_{i=1}^{n} \frac{\phi\{y_i - \mu_i\}}{V_i(\mu_i)}. \end{aligned} \tag{8.3}$$

It is easy to show that the elements of $U(\boldsymbol{\mu}|\phi)$, which are first derivatives of

$Q(\boldsymbol{\mu}|\phi)$, have the following properties:

$$\begin{aligned} E\{U_i(\mu_i|\phi)\} &= 0, \\ var\{U_i(\boldsymbol{\mu}_i|\phi)\} &= \frac{\phi}{V_i(\mu_i)} \\ -E\left\{\frac{\partial}{\partial \mu_i} U_i(\mu_i|\phi)\right\} &= \frac{\phi}{V_i(\mu_i)} \end{aligned}$$

Notice that, given the first, the third of these properties constitute an condition that is analogous to the condition in regular likelihood problems that the expected information is equal to the negative expectation of second derivatives of the log likelihood. In addition, the first property given above implies that

$$\begin{aligned} E\left\{\frac{\partial Q_i(\mu_i|\phi)}{\partial \beta_k}\right\} &= E\left\{\frac{\partial Q_i(\mu_i|\phi)}{\partial \mu_i}\frac{\partial \mu_i}{\partial \beta_k}\right\} \\ &= E\left\{U_i(\mu_i|\phi)\frac{\partial \mu_i}{\partial \beta_k}\right\} = 0. \end{aligned} \tag{8.4}$$

Quasi-likelihood functions share the property given in expression (8.4) with true likelihood functions, and this then suggests that maximum quasi-likelihood estimates of $\boldsymbol{\beta}$ might be found by solving these equations (absent the expectation operator) for the elements of $\boldsymbol{\beta}$.

In a similar way, the second and third properties imply that,

$$\begin{aligned} E\left\{\frac{\partial Q_i(\mu_i|\phi)}{\partial \beta_k}\frac{\partial Q_i(\mu_i|\phi)}{\partial \beta_j}\right\} &= E\left\{\{U_i(\mu_i|\phi)\}^2\frac{\partial \mu_i}{\partial \beta_k}\frac{\partial \mu_i}{\partial \beta_j}\right\} \\ &= -E\left\{\frac{\partial U_i(\mu_i|\phi)}{\partial \mu_i}\frac{\partial \mu_i}{\partial \beta_k}\frac{\partial \mu_i}{\partial \beta_j}\right\} \\ &= \frac{\phi}{V_i(\mu_i)}\frac{\partial \mu_i}{\partial \beta_k}\frac{\partial \mu_i}{\partial \beta_j}. \end{aligned}$$

Given these results, we may derive a analogy to a Fisher scoring algorithm in a manner similar to that used for maximum likelihood estimation of $\boldsymbol{\beta}$ in basic generalized linear models. Now, however, we are maximizing the quasi-likelihood function rather than the log likelihood function. The result is an algorithm that updates parameter estimates as,

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + \delta\boldsymbol{\beta}^{(m)},$$

where,

$$\delta\boldsymbol{\beta}^{(m)} = \left(\boldsymbol{D}^T\boldsymbol{V}^{-1}\boldsymbol{D}\right)^{-1}\boldsymbol{D}^T\boldsymbol{V}^{-1}(\boldsymbol{y}-\boldsymbol{\mu})\Big|_{\boldsymbol{\beta}=\boldsymbol{\beta}^{(m)}}. \tag{8.5}$$

In (8.5) $\boldsymbol{V}$ is the $n \times n$ diagonal matrix with entries $V_i(\mu_i)$ and $\boldsymbol{D}$ is the $n \times p$ matrix with ik^{th} element $\partial\mu_i/\partial\beta_k$; recall that p is the dimension of $\boldsymbol{\beta}$. Notice that, in this algorithm, the parameter ϕ has canceled, in the same way that it did for development of the Fisher scoring algorithm for standard generalized linear models.

Inference for maximum quasi-likelihood is based on a result that if $\tilde{\boldsymbol{\beta}}$ denotes the maximum quasi-likelihood estimator of $\boldsymbol{\beta}$, then:

(i) $\tilde{\boldsymbol{\beta}}$ is consistent for $\boldsymbol{\beta}$.

(ii) $\sqrt{n}(\tilde{\boldsymbol{\beta}}-\boldsymbol{\beta})$ is asymptotically normal with mean $\mathbf{0}$ and covariance matrix $(n/\phi)\left(\boldsymbol{D}^T\boldsymbol{V}^{-1}\boldsymbol{D}\right)^{-1}$.

Given the asymptotic normality of $\tilde{\boldsymbol{\beta}}$, inference for $\boldsymbol{\beta}$ can be based on normal theory forms with the asymptotic covariance matrix

$$cov(\tilde{\boldsymbol{\beta}}) = \frac{1}{\phi}\left(\boldsymbol{D}^T\boldsymbol{V}^{-1}\boldsymbol{D}\right)^{-1}. \tag{8.6}$$

Use of (8.6) in practice requires an estimate of ϕ. A common estimator is the moment-based estimator,

$$\hat{\phi} = \left[\frac{1}{n-p}\sum_{i=1}^{n}\frac{\{y_i-\hat{\mu}_i\}^2}{V(\hat{\mu}_i)}\right]^{-1}.$$

When might one consider the use of quasi-likelihood in estimation and inference? One situation in which quasi-likelihood presents a viable option for estimation and inference is if a standard generalized linear model has been used, but the resulting fit exhibits a less than totally adequate description of the observed variances. That is, we formulate a model with $Y_1, \ldots, Y_n$ distributed according to an exponential dispersion family, which then dictates the variance function $V(\mu_i)$. Diagnostics, such as described in McCullagh and Nelder (1989) may reveal that this assumed variance function does not describe the data in a completely satisfactory manner. We may then choose to envisage a model with essentially the same type of random component only with a modified variance function.

Example 8.2

McCullagh and Nelder (1989, Chapter 9.2.4) present an example in which the initial model was a binomial random component with logistic link, and hence variance function $V(\mu_i) = \mu_i(1 - \mu_i)$ (the response variables Y_i were taken as proportions rather than counts). Diagnostic plots (McCullagh and Nelder 1989, p. 331) showed that the variances appeared to decrease too rapidly (as a function of μ_i) at the extremes (for μ_i close to 0 and 1). They then suggested a variance function $V(\mu_i) = \mu_i^2(1 - \mu_i)^2$ as an alternative. This variance function does not correspond to any of the standard exponential dispersion family distributions and, hence, a fully specified random component is no longer available. A quasi-likelihood analysis was then conducted of these data with

$$E(Y_i) = \mu_i; \qquad var(Y_i) = \frac{1}{\phi}V(\mu_i) = \frac{1}{\phi}\mu_i^2(1 - \mu_i)^2,$$

and with $\log\{\mu_i/(1-\mu_i)\} = \boldsymbol{x}_i^T\boldsymbol{\beta}$. This then leads to the quasi-likelihood (in μ_i),

$$Q_i(\mu_i|\phi) = (2y_i - 1)\log\left(\frac{\mu_i}{1-\mu_i}\right) - \frac{y_i}{\mu_i} - \frac{1-y_i}{1-\mu_i}.$$

The potential drawbacks to quasi-likelihood are relatively obvious, and include:

1. The asymptotic result given for maximum quasi-likelihood estimators in this section was presented as analogous to *part* of the result of Likelihood Theorem 2 in Chapter 5 of the Stat 520 notes. The part that is missing in the quasi-likelihood result is the statement of asymptotic efficiency (part (iii) of Likelihood Theorem 2).

2. While inferential procedures are available for parameters in the systematic model component, represented in this section as $\boldsymbol{\beta}$, quasi-likelihood no longer offers a vehicle by which to make inferences about any other portion of the model such as quantiles or other functionals of the distribution.

3. Related to item 2, interpretation of results relative to the underlying scientific phenomenon or mechanism of interest becomes less well-defined. Consider Example 8.2 given previously. By replacing the variance function of a binomial with something else, we have admitted that we do not understand the observation process, since we no longer have an actual model. This should (in my opinion) cast substantial doubt on whether we have much of a grasp on modeling the scientific phenomenon of interest through the systematic model component.

Drawing on the third comment above, quasi-likelihood appears in many cases to be an attempt to account for the additional variance in an observable

process without having to go to the trouble of modeling it in an adequate manner; effort is diverted to estimation rather than modeling.

8.1.3 Extended Quasi-Likelihood

It was noted immediately following Example 8.1 that, in exponential dispersion family situations, the additive constant that separates a quasi-likelihood and the true log likelihood will depend on both $\boldsymbol{y}$ and ϕ, and we have been careful to write the quasi-likelihood and quasi-score functions as conditional on the dispersion parameter ϕ. With this conditioning, the quasi-likelihood method produces estimates for systematic model component parameters that behave in a manner similar to that of maximum likelihood estimates, minus asymptotic efficiency. If, however, we would like a quasi-likelihood function that is a sort-of likelihood in terms of ϕ as well as in terms of μ_i; $\quad i = 1, \ldots, n$, something else is needed than what has been developed to this point. That development is the intent of what is typically called *extended quasi-likelihood* (Nelder and Pregibon 1987).

Extended quasi-likelihood functions are derived in McCullagh and Nelder (1989) by essentially pre-supposing the end product. Barndorff-Nielsen and Cox (1989) give a progression in which extended quasi-likelihood is derived as a *tilted Edgeworth* expansion (also often called a saddlepoint approximation). In either case, what results is the extended quasi-likelihood of Nelder and Pregibon (1987), written for a single random variable Y_i as,

$$\begin{aligned} Q_i^+(\mu_i, \phi) &= \int_{y_i}^{\mu_i} \frac{\phi\{y_i - t\}}{V(t)}\, dt - \frac{1}{2}\log\{2\pi(1/\phi)V(y_i)\} \\ &= Q_i(\mu_i, \phi) - \frac{1}{2}\log\{2\pi(1/\phi)V(y_i)\}. \end{aligned} \tag{8.7}$$

The only difference between $Q_i(\mu_i|\phi)$ in (8.2) and what has been written as $Q_i^+(\mu_i, \phi)$ in (8.7) is whether ϕ is considered a fixed value in the function, or part of the argument. As pointed out by McCullagh and Nelder (1989, p. 350) the justification of extended quasi-likelihood as a saddlepoint approximation depends on some type of assumption that renders the contribution of higher-order cumulants to the Edgeworth expansion used negligible. Typically, in such an expansion for a density, cumulants higher than order 4 are dropped (e.g., Stuart and Ord 1987); in derivation of extended quasi-likelihood we make this assumption for cumulants of greater than order 2.

The use of extended quasi-likelihood is perhaps epitomized by models of the generalized linear model form in which we attempt to model both the expectation and variance as functions of covariates. Note that this is the generalized linear model counterpart of additive error models with variance that depends on unknown parameters. One way to specify such models is given in McCullagh and Nelder (1989, Chapter 10.2) as,

$$E(Y_i) = \mu_i; \qquad g(\mu_i) = \eta_i = \boldsymbol{x}_i^T \boldsymbol{\beta};$$

$$var(Y_i) = \frac{1}{\phi_i} V(\mu_i),$$

and,

$$\phi_i = E\{d_i(Y_i, \mu_i)\}; \qquad h(\phi_i) = \boldsymbol{u}_i^T \boldsymbol{\gamma};$$

$$var\{d_i(Y_i, \mu_i)\} = \frac{1}{\tau}\phi^2.$$

In the above model formulation, $d_i(Y_i, \mu_i)$ is chosen as a measure of dispersion, often,

$$d_i(Y_i, \mu_i) = \frac{(Y_i - \mu_i)^2}{V(\mu_i)}.$$

The function $h(\cdot)$ is a "dispersion link function" that connects the expected value of $d_i(Y_i, \mu_i)$, namely ϕ_i, with covariates $\boldsymbol{u}_i$ (often some or all of $\boldsymbol{x}_i$, just

as in additive error models), and the relation between ϕ and $var\{d_i(Y_i, \mu_i)\}$ is dictated if one uses extended quasi-likelihood for estimation. McCullagh and Nelder (1989, Chapter 10.5) give several additional adjustments to the extended quasi-likelihood procedure that may be desirable.

8.2 Generalized Estimating Equations

Thus far in this chapter our discussion has been entirely in the context of independent response variables Y_i; $i = 1, \ldots, n$. One of the primary areas of application for quasi-likelihood and similar ideas, however, has certainly been longitudinal studies in which random variables correspond to repeated observation of particular sampling units or individuals (e.g., Zeger and Liang 1986; Liang and Zeger 1986; Zeger, Liang, and Albert 1988). This type of situation is often dealt with using a model that implies correlated random variables within individuals. A fundamental property of this setting is that we do, in fact, have independent realizations across groups or individuals of some type of marginal model depending on the same parameters.

In the dependence setting, quasi-likelihood blends into what we will call *Estimating Functions* because the starting point is really the quasi-score rather than the quasi-likelihood. In fact, the quasi-likelihood itself is rarely mentioned or defined (but see McCullagh and Nelder 1989, Chapter 9.3.2). The term *Generalized Estimating Equations* has also been used, notably by Zeger and Liang (1986). The fundamental concept follows from noting two things about what was presented previously as the quasi-score function, which may be written, for $k = 1, \ldots, p$, as,

$$\sum_{i=1}^{n} U_i(\mu_i|\phi)\frac{\partial \mu_i}{\partial \beta_k} = \sum_{i=1}^{n} \frac{\phi\{y_i - \mu_i\}}{V_i(\mu_i)}\frac{\partial \mu_i}{\partial \beta_k}.$$

Note that, again, the roots of these equations in $\boldsymbol{\beta}$ will not involve ϕ. Secondly, note that we could use the same form of these equations with y_i and μ_i replaced by vectors $\boldsymbol{y}_i \equiv (y_{i,1}, \ldots, y_{i,m_i})^T$ and $\boldsymbol{\mu}_i \equiv (\mu_{i,1}, \ldots, \mu_{i,m_i})^T$, and $V_i(\mu_i)$ replaced by an $m_i \times m_i$ matrix $\boldsymbol{V}_i$ which gives, up to the scalar multiple ϕ, the covariance matrix of $\boldsymbol{Y}_i$. Let n now denote the number of independent vectors $\{\boldsymbol{y}_i : i = 1, \ldots, n\}$ (e.g., number of individuals in a longitudinal study). The above equations may then be written as what Zeger and Liang (1986) called generalized estimating equations,

$$\sum_{i=1}^{n} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{S}_i = 0, \tag{8.8}$$

where $\boldsymbol{S}_i \equiv (\boldsymbol{y}_i - \boldsymbol{\mu}_i)$ is $m_i \times 1$, $\boldsymbol{D}_i$ is $m_i \times p$ with jr^{th} element $\partial\mu_{i,j}/\partial\beta_r$ and $\boldsymbol{V}_i$ is $m_i \times m_i$ with structure to be given directly. For a particular model, an appropriate structure must be chosen for $\boldsymbol{V}_i$; $i = 1, \ldots, n$. Liang and Zeger (1986) suggested using a *working correlation matrix* $\boldsymbol{R}_i(\alpha)$ to help define $\boldsymbol{V}_i$ as,

$$\boldsymbol{V}_i = \boldsymbol{A}_i^{1/2} \boldsymbol{R}_i(\alpha) \boldsymbol{A}_i^{1/2}, \tag{8.9}$$

where $\boldsymbol{A}_i$ is diagonal with elements given by what, in the independence case were the variance functions $V(\mu_i)$ and are now functions $w(\mu_{i,j})$ such that $var(Y_{i,j}) \propto w(\mu_{i,j})$.

Estimates of $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T$ are then found by solving the estimating equations (8.8) with the definition of $\boldsymbol{V}_i$ as in (8.9). In order to do this, a value is needed for the correlation matrix parameter α. We will discuss this, and estimation of ϕ shortly. But first, it can be helpful to describe the types of matrices $R_i(\alpha)$ that might be used in some structures.

1. Independence.

   If $R_i(\alpha) = I_{m_i}$, then the estimating equations (8.8) reduce to those

of quasi-score functions in the independence case. Nevertheless, this choice of $R(\alpha)$ is sometimes used to obtain starting values for iterative algorithms to solve the full equations (8.8) with a different choice of $R(\alpha)$.

2. Random Effects Structure.

$$R_i(\alpha) = \begin{pmatrix} 1 & \alpha & \dots & \alpha \\ \alpha & 1 & \dots & \alpha \\ \vdots & \vdots & \dots & \vdots \\ \alpha & \alpha & \dots & 1 \end{pmatrix}_{m_i \times m_i} .$$

This is the correlation structure of a linear random effects model in which all variables in the same group or within the same individual (here the variables contained in $\boldsymbol{Y}_i$) have the same correlation.

3. Autoregressive Process of Order k.

Here, we would take the uv^{th} element of the matrix $\boldsymbol{R}_i$ to be

$$[\boldsymbol{R}_i]_{u,v} = \begin{cases} \alpha^{|t_{i,u} - t_{i,v}|}, & |t_{i,u} - t_{i,v}| \leq k \\ 0 & |t_{i,u} - t_{i,v}| > k, \end{cases}$$

where $t_{i,u}$ and $t_{i,v}$ are the uth and v^{th} observation times of the group or individual indexed by i.

For estimation of $\boldsymbol{\beta}$, α, and ϕ, the general prescription is to utilize a Gauss-Newton type of algorithm in which one iteration for $\boldsymbol{\beta}$ is conducted using current estimates of α as fixed, then estimating new α for the current β and so forth. At each iteration, α is updated via a moment-based estimator, and ϕ is also estimated based on a moment-type estimator. We will not present the details here (since we've covered the basic ideas already), but see Zeger and Liang (1986) or Pawitan (2001) for all the particulars.

Inference about $\boldsymbol{\beta}$ is made using asymptotic normality of the GEE estimates. The form of the asymptotic covariance matrix for $\hat{\boldsymbol{\beta}}$ is given in Zeger and Liang (1986) and Liang and Zeger (1986), and is estimated by using moment-based estimators $\hat{\alpha}$ and $\hat{\phi}$ as plug-in values.

8.3 Estimating Functions

The quasi-score function and generalized estimating equations of the previous sections are special cases of what are known as *estimating functions*. Given random variables $\boldsymbol{Y} \equiv (Y_1, \ldots, Y_n)^T$ with a density $f_Y(\boldsymbol{y}|\boldsymbol{\theta})$, an unbiased estimating function for $\boldsymbol{\theta}$ can be defined quite simply as any function $v(\boldsymbol{Y}, \boldsymbol{\theta})$ such that, for all values of $\boldsymbol{\theta} \in \Theta$,

$$E\{v(\boldsymbol{Y}, \boldsymbol{\theta})\} = 0. \tag{8.10}$$

If we are interested in only a part of $\boldsymbol{\theta}$, then the estimating function is defined relative to only that component or components.

Example 8.3

Let $Y_1, \ldots, Y_n$ be *iid* $N(\mu, \sigma^2)$. Possible estimating functions for μ include,

$$\begin{aligned} v(\boldsymbol{y}, \mu) &= \sum_{i=1}^{n} (y_i - \mu) = 0, \\ v(\boldsymbol{y}, \mu) &= \sum_{i=1}^{n} sgn(y_i - \mu) = 0, \\ v(\boldsymbol{y}, \mu) &= \sum_{i=1}^{n} \psi(y_i - \mu) = 0, \end{aligned}$$

where in this third possibility, for some real number r,

$$\psi(x) = x(r^2 - x^2)^2 I(-r < x < r).$$

Estimates resulting from solving these three possible estimating functions for μ are the sample mean, the sample median, and a robust estimator that corresponds to use of the function $\psi(\cdot)$ which in the above is known as the *biweight* or *bisquare* function, attributed to Tukey (e.g., Hampel *et al.* 1986).

Asymptotic inference for estimating functions is developed in much the same way as for maximum likelihood estimators and, although we did not cover it in any detail, maximum quasi-likelihood estimators. Specifically, in the case of a scalar parameter θ, suppose conditions sufficient to show the following are assumed.

1. The solution to an unbiased estimating function $\tilde{\theta}$ may be expanded as

$$v(\boldsymbol{y}, \tilde{\theta}) \approx v(\boldsymbol{y}, \theta) + \left. \frac{\partial v(\boldsymbol{y}, \tilde{\theta})}{\partial \tilde{\theta}} \right|_{\tilde{\theta}=\theta} (\tilde{\theta} - \theta),$$

   from which we have, since $v(\boldsymbol{y}, \tilde{\theta}) = 0$ by definition,

$$\tilde{\theta} - \theta = -v(\boldsymbol{y}, \theta) \left\{ \left. \frac{\partial v(\boldsymbol{y}, \tilde{\theta})}{\partial \tilde{\theta}} \right|_{\tilde{\theta}=\theta} \right\}^{-1}.$$

2. The random version of $v(\cdot)$ satisfies,

$$n^{-1/2} v(\boldsymbol{Y}, \theta) \quad \text{is} \quad AN(0, q(\theta)),$$

   for some function $q(\cdot)$.

3. The random version of the derivative of $v(\cdot)$ satisfies,

$$\frac{1}{n} \frac{\partial v(\boldsymbol{Y}, \theta)}{\partial \theta} \xrightarrow{p} s(\theta),$$

   for some function $s(\cdot)$.

Then the result is (Barndorff-Nielsen and Cox 1994, p.303),

$$\tilde{\theta} \quad \text{is} \quad AN\left(\theta, \frac{q(\theta)}{n\, s^2(\theta)}\right).$$

There appear to be two major areas in which estimating functions surface. The first is an an alternative theory of estimation, in which estimating functions are presented as an alternative to both least squares (viewed as an exact theory procedure) and maximum likelihood. This is, for example, the view offered by Godambe and Kale (1991, p. 17) in which the authors claim to demonstrate that estimating functions "unifies the method of maximum likelihood and the method of minimum variance unbiased estimation in the case of parametric models". Under this viewpoint we must deal with the development of criteria under which one of a variety of possible estimating functions can be deemed optimal. There will be any number of unbiased estimating functions that can be developed in most situations (as illustrated in Example 8.3). A variety of possible optimality criteria are offered by Godambe and Heyde (1987). A slightly different view is offered by Barndorff-Nielsen and Cox (1994) who point out that, in the case of $Y_1, \ldots, Y_n \sim iid$ with common density $f(y|\theta)$ for a scalar θ, the two matrices $q(\cdot)$ and $s(\cdot)$ in conditions 2 and 3 listed above for asymptotic normality of $\tilde{\theta}$ become

$$q(\theta) = var\left\{\frac{\partial \log\{f(y_i|\theta)}{\partial\theta}\right\},$$

$$s(\theta) = E\left\{\frac{\partial^2 \log\{f(y_i|\theta)}{\partial\theta^2}\right\},$$

so that $q(\theta) = -s(\theta)$ and the asymptotic result leads to the solution to the likelihood equation (an estimating function) being asymptotically normal with variance given by the inverse information. This then motivates them to suggest a criterion

$$\rho_v(\theta) = \frac{[E\{\partial v(\boldsymbol{Y},\theta)/\partial\theta\}]^2}{var\{v(\boldsymbol{Y},\theta)\}},$$

as a measure of the lost efficiency of the solution to any estimating function $v(\boldsymbol{Y},\theta)$; this is because $\rho_v(\theta)$ is maximized by the likelihood score function.

The second major area in which estimating functions play a role is in the development of robust estimators, as indicated by the third possible estimating function given in Example 8.3 for the one-sample normal problem. This is the flavor of the presentation by, for example, Pawitan (2001, Chapter 14). In this context, estimating functions are also often called M-estimators. See, for example, Hampel *et. al.* (1986), and Carroll and Ruppert (1988, Chapter 7).

One aspect of estimating functions that deserves mention is that, if $v_1(\cdot), \ldots, v_p(\cdot)$ are unbiased estimating functions, then any linear combination, $\sum a_j v_j$ is also an unbiased estimating function. This property can sometimes be utilized to form simple estimating functions in even complex situations. For example, in the autoregressive process

$$Y(t) = \theta Y(t-1) + \epsilon(t),$$

where $\epsilon(t) \sim iid\, N(0,1)$, we could take estimating functions,

$$v_t = y(t) - \theta y(t-1),$$

and form the combination,

$$v(\boldsymbol{y}, \theta) = \sum_t y(t-1)\, v_t.$$

An example of such development is given in McCullagh and Nelder (1989, p.341).